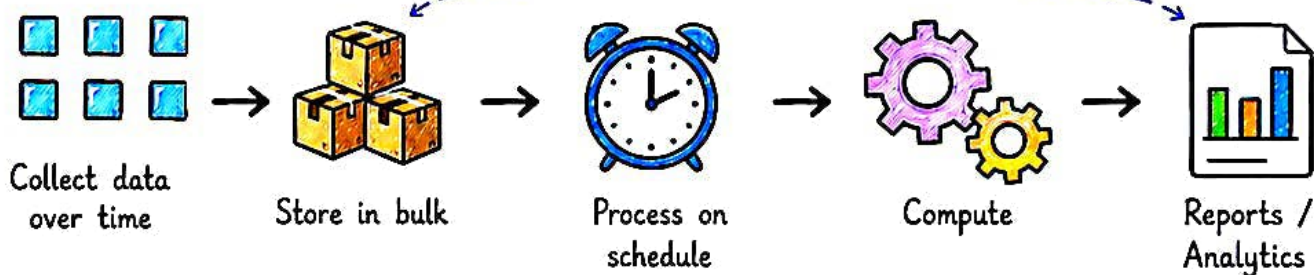


BATCH vs STREAM PROCESSING

Should your data **WAIT** ⌚ or move in **REAL TIME**? ⚡

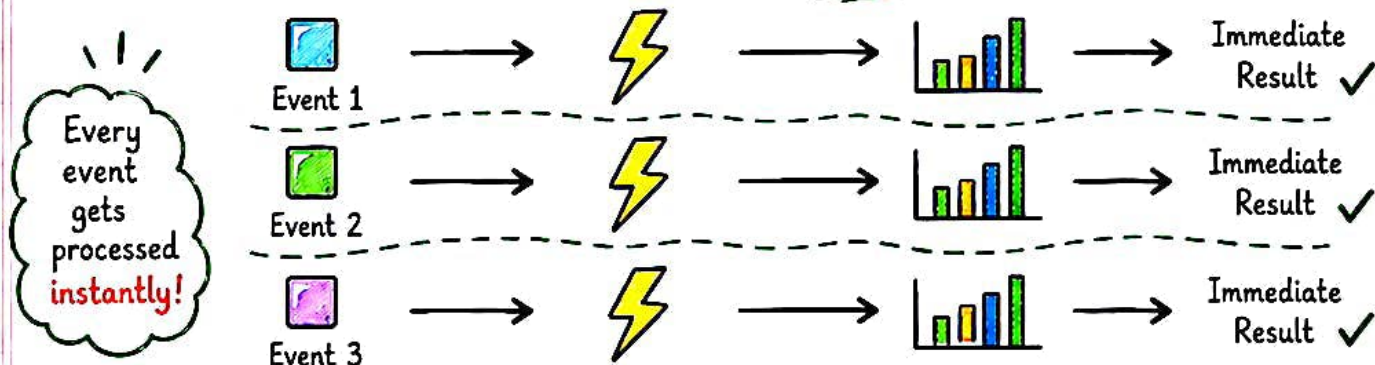
BATCH PROCESSING

Waits for all data
then processes



STREAM PROCESSING

Processes data as it arrives
(real-time)



Two ways to process data.
Two **very different** use cases.



BATCH PROCESSING

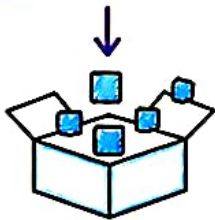
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Collect data first → Process later

HOW IT WORKS



DATA SOURCES



COLLECT DATA

Accumulate over
a period of time



SCHEDULED JOB

Run at fixed intervals
(e.g., daily, hourly)



PROCESS

Clean, transform
and aggregate data



WAREHOUSE / REPORTS

Store in data warehouse
and generate reports

EXAMPLES

- ✓ Daily sales report
- ✓ Monthly billing
- ✓ Payroll processing
- ✓ Inventory summary
- ✓ Nightly ETL jobs

KEY CHARACTERISTICS

- Processes large volume of data at once
- Higher latency (minutes to hours)
- Cost effective for large data sets
- Simple to design and operate



THINK:

Large amount of data processed together



STREAM PROCESSING

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Process data continuously **as it arrives**

HOW IT WORKS



EVENTS ARRIVE
Data is generated continuously



INGEST IN REAL-TIME
Events are captured as they happen



PROCESS IMMEDIATELY
Analyze, enrich and apply logic on the fly



REAL-TIME OUTPUT
Instant insights, alerts and dashboards

EXAMPLES

- ✓ Fraud detection
- ✓ Live dashboards
- ✓ IoT monitoring
- ✓ Website clickstream
- ✓ Real-time alerts
- ✓ Live order tracking

KEY CHARACTERISTICS

- Processes data **one event at a time**
- **Low latency** (seconds or milliseconds)
- Continuous and **unbounded data flow**
- **Always on** and real-time

COMMON STREAM PROCESSING TOOLS



Kafka

Real-time data ingestion & streaming



Spark

Stream processing with Spark Streaming



Flink

Stateful stream processing engine



Kinesis

Managed real-time streaming by AWS



Airflow

Orchestrate real-time pipelines



THINK: Event **arrives** → Process **now** → Get result **immediately!**



BATCH vs STREAM PROCESSING

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Key differences at a glance



ASPECT	BATCH PROCESSING 	STREAM PROCESSING 
 Data Arrival	Data is collected over time and processed in groups	Data arrives continuously as individual events
 Processing Style	Scheduled / Periodic (e.g., hourly, daily)	Continuous / Real-time (24 x 7)
 Latency	High latency (minutes to hours)	Very low latency (seconds to milliseconds)
 Data Volume	Handles large volumes at a time	Handles small events continuously
 Use Case	Historical analysis, reporting, ETL jobs	Real-time analytics, alerts, monitoring
 Complexity	Generally simpler to design & operate	More complex to design & operate
 Cost Pattern	Lower infrastructure cost (process less frequently)	Higher infrastructure cost (always running)
 Example	Daily sales report, monthly billing, payroll	Fraud detection, live dashboard, IoT data

BATCH IN A NUTSHELL

- Wait for data to accumulate
- Process in bulk
- Great for historical insights
- Works on schedules



STREAM IN A NUTSHELL

- Process data as it flows
- Low latency
- Great for real-time actions
- Always on and continuous



REMEMBER:

Batch looks at the PAST.

Stream acts in the PRESENT.



REAL-WORLD EXAMPLE

E-COMMERCE PLATFORM

5/6

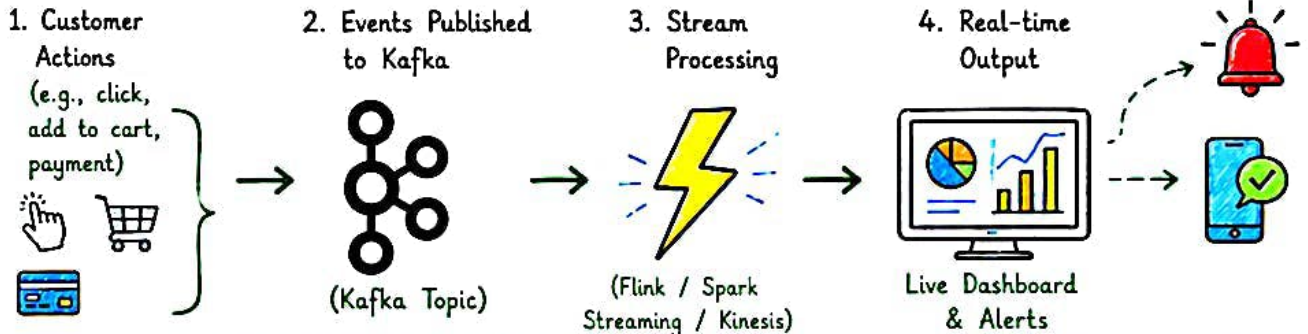
Same business, **different** processing needs!

BATCH PROCESSING (Daily Sales Report)



★ Use Case: Management dashboard, daily reports, inventory summary

STREAM PROCESSING (Real-time Insights)



★ Use Case: Fraud detection, live dashboard, stock updates, real-time notifications

TOOLS YOU MUST KNOW

APACHE
Spark

Batch &
Stream
Processing

kafka

Event
Streaming

Flink

Real-time
Stream
Processing

amazon
KINESIS

Managed
Streaming
Service

Apache
Airflow

Workflow
Orchestration
(Batch Jobs)



REMEMBER:

Choose the **right** processing for the **right use case!**



WHEN SHOULD YOU USE WHICH?

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CHOOSE BATCH



WHEN:

- ✓ Real-time results are not required
- ✓ You are working with historical data
- ✓ Data is generated in large volumes at intervals
- ✓ Scheduled jobs and periodic reports
- ✓ Simpler processing is acceptable

EXAMPLES

- Daily sales report
- Monthly billing
- Payroll processing
- Inventory summary



CHOOSE STREAM



WHEN:

- ✓ Seconds or milliseconds matter
- ✓ Continuous events are generated
- ✓ Real-time alerts are critical
- ✓ Live dashboards & monitoring are needed
- ✓ You need instant reaction to events

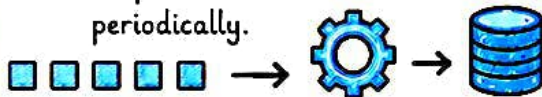
EXAMPLES

- Fraud detection
- Live analytics dashboard
- IoT device monitoring
- Website clickstream
- Real-time order tracking



INTERVIEW TAKEAWAY

Batch processes bounded groups of data periodically.



Stream processes continuous events as they arrive with low latency.



Good Data Engineers understand when to use BOTH.



SAVE THIS

for your Data Engineering learning & interview preparation!



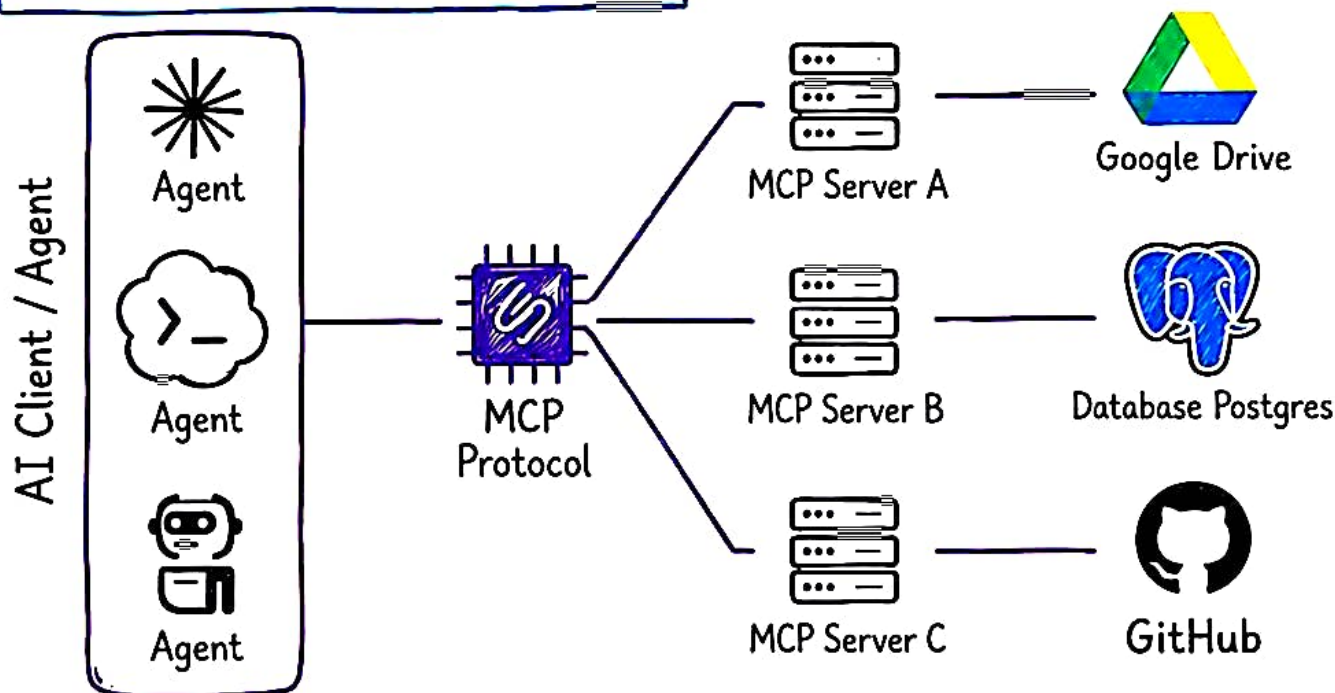
Comment "DATA"

if you found this helpful! ❤️

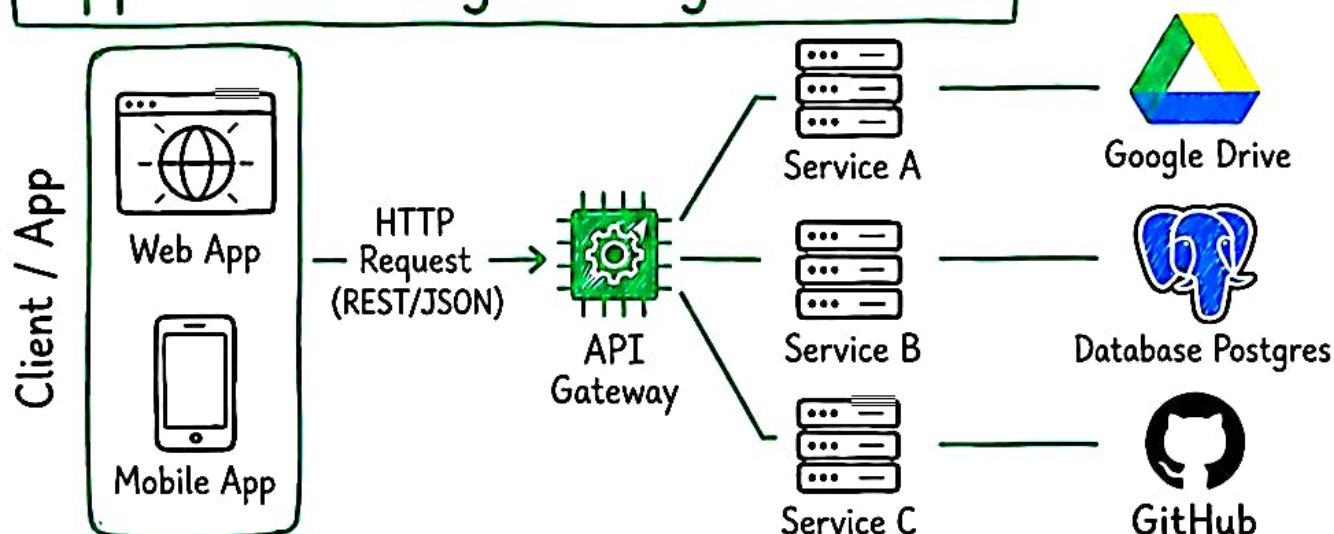


MCP vs API

Model Context Protocol



Application Programming Interface



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